

Application Code & Replication Benchmark Suite

Technical Documentation & MATLAB–R Bridge Setup Guide

Replication Package for:

” _____ ”

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1 Overview & Repository Structure

This repository contains every single routine and the exact datasets required to replicate all empirical applications and simulations presented in the manuscript.

All source code files (.m and .R) are self-documented. Detailed mathematical definitions, input/output specifications, diagnostic algorithms, and option flags are located directly within the header comment blocks of each individual script.

2 MATLAB–R Bridge Setup Guide

When analyzing **ordinal, Likert-scale, or mixed continuous-ordinal data**, the MATLAB function `processData.m` delegates the computation of polychoric, tetrachoric, and polyserial correlation matrices to R using the `psych` package.

Follow the steps below to configure the local bridge environment on your system.

Operating System Compatibility Note:

The automatic detection of R's binary path is optimized for **Windows** systems. macOS and Linux users can execute the bridge seamlessly by ensuring that **Rscript** is available in their system's global PATH.

2.1 Prerequisites & Dependencies

1. **MATLAB:** R2018b or newer recommended.
2. **R Environment:** Download and install the latest version from CRAN (<https://cran.r-project.org/>).
3. **Required R Packages:** Open R or RStudio and run:

```
install.packages(c("psych", "R.matlab"))
```

2.2 Directory Architecture Requirements

The pipeline uses a fixed file exchange directory inside your system's **Documents** directory to exchange data frames between MATLAB and R.

1. Locate system Documents directory:

- **Windows:** `C:\Users\<YourUsername>\Documents\`
- **macOS:** `/Users/<YourUsername>/Documents/`
- **Linux:** `/home/<YourUsername>/Documents/`

2. Create exchange subfolder: Create the path `rfiles/matlab_bridge/` inside your Documents directory.

Full Windows Example: `C:\Users\<YourUsername>\Documents\rfiles\matlab_bridge\`

3. Position the R script: Place the script file `calc_polychoric.R` inside that folder:

`.../Documents/rfiles/matlab_bridge/calc_polychoric.R`

2.3 Execution Workflow

When calling `processData(X, Q, opts)` in MATLAB with `opts.is_ordinal = 1`, the system performs the following sequence automatically:

```
[MATLAB] Exports matrix X --> temp_X_data.mat
          |
          v
          [Rscript Executed via System Call]
          calc_polychoric.R (psych::polychoric / mixedCor)
          |
          v
[MATLAB] Imports matrix R <-- temp_R_poly.mat
```

1. MATLAB writes data matrix X to `temp_X_data.mat`.
2. MATLAB executes `calc_polychoric.R` via system shell call using `Rscript`.
3. R classifies variables (e.g., maximum integer values ≤ 7 as ordinal items) and computes the correlation matrix R .
4. R saves output metadata to `temp_R_poly.mat`.
5. MATLAB loads R , deletes temporary exchange files, and proceeds with Factor Analysis.

3 Troubleshooting & Fallback Mechanisms

- **Custom R Paths (Windows):** If R is installed outside default paths (`C:\Program Files\R\`), add R's `bin` directory to your system's `PATH` environment variable.
- **Terminal Test (macOS/Linux):** Verify accessibility by entering `Rscript --version` in your terminal.
- **Automatic Fallback Guardrail:** If the bridge call fails, lacks dependencies, or optimization does not converge, the pipeline generates a warning message and automatically falls back to standard Pearson pairwise correlations.